

Statement of Work

National Institutes of Health (NIH) National Cancer Institute (NCI) Center for Biomedical Informatics and Information Technology (CBIIT) Biomedical Information Technology, Software Development, and Informatics Support (BITSDIS 2)

2.1. PURPOSE

The purpose of this Indefinite-Delivery Indefinite-Quantity (IDIQ) vehicle is to acquire professional biomedical informatics, data science, information technology, software development, and artificial intelligence support services in support of the National Institutes of Health (NIH), National Cancer Institute (NCI) to assist in achieving NIH's mission. NCI's Center for Biomedical Informatics and Information Technology (CBIIT) is the sponsor of this requirement; however task orders may be issued on behalf of any NCI Division, Office, or Center (DOC) in addition to any NIH Institute or Center (IC).

2.2. BACKGROUND

The NIH's mission is to seek fundamental knowledge about the nature and behavior of living systems and to apply that knowledge to enhance health, lengthen life, and reduce illness and disability. The NCI is one of the 27 institutes that make up the NIH, the nation's primary federal agency for biomedical and public health research. that conducts and supports research, training, and health information dissemination with respect to the cause, prevention, diagnosis, and treatment of cancer, rehabilitation from cancer, and the continuing care of cancer patients and the families of cancer patients.

CBIIT is an important NCI component entrusted to lead and coordinate efforts in the fields of informatics, data science, and information technology. CBIIT is defining, designing, developing, and deploying key infrastructure, standards, and application-level components of the Information Technology (IT) fabric to support a modern vision for integrated cancer and biomedical research, translational and population science, and personalized medicine. CBIIT establishes and develops the framework, infrastructure, standards, processes, and tools for data collection and management to ensure the long-term value of NCI's investment in research through data accessibility, interoperability, sharing and reuse. CBIIT creates mechanisms to build and support information networks enabling all constituencies in the cancer community – NCI programs, other institutions, researchers, physicians, and patients – to facilitate new discovery, integrate data, implement knowledge, make research results available for the community, minimize the burden of cancer, and improve the lives of patients.

Major ongoing NCI initiatives and programs supported under this vehicle include, but are not limited to, the Cancer Research Data Commons (CRDC), a comprehensive Clinical Reporting and Research Informatics (CRRI) initiative that include elements of the Cancer Moonshot and Precision Medicine Initiative (PMI). CRRI will include Precision Medicine Activities (PMA), Correlative Study Management (CSMS), the Clinical Trials Reporting Program (CTRP) and CTRP's Clinical Trials Reporting Office (CTRO), as well as Computational Genomics and

Bioinformatics research and ancillary support services. Specifics include:

- A Precision Medicine Analysis and Coordination Center (PMACC) that will support current and next generation NCI precision medicine trials. This center will provide support for existing trials, new treatment studies for the existing trials as well as new major initiatives; each of which will be composed of an umbrella/basket screening platform and linked clinical trials. Each master protocol will require both a MATCHBox precision medicine trial platform to determine patient eligibility for specific therapies and associated precision diagnostic laboratories provided by the NCI/DCTD Molecular and Immunologic Diagnostics Network of Laboratories (MDNet). The PMACC will provide the treatment assignment decisions based on data obtained from MDNet, other qualified laboratories, and the trial groups. The MDNet will provide primary diagnostic screening for trial arms that require specialized laboratory testing for patient allocation,
- CIDC (Cancer Immunologic Data Center) and CIMAC (Cancer Immune Monitoring and Analysis Centers) are both components of a collaborative effort aimed at advancing cancer immunotherapy research. CIDC is the data management and analysis platform that works alongside CIMAC. It collects, organizes, and stores the data generated by the CIMAC centers and other cancer immunotherapy trials. It also facilitates data sharing among researchers, ensuring that data is accessible for analysis and fostering collaboration. Supports the use of advanced computational tools and analytics to identify patterns in immune responses and potential biomarkers for treatment success. The Correlative Study Management System (CSMS) supports the NCI Cancer Immune Monitoring and Analysis Centers (CIMAC) Biomarker Network and works along the CIDC to support tracking specimen from enrolling sites. CBIIT has migrated the CIDC to the NCI Cloud and both CIDC and CSMS should be maintained and enhanced to support study management dashboards, unique sample identifier generation, manifest and sample label generation, biospecimen tracking, document management and information management capabilities.
- The Clinical Trial Reporting Program (CTRP) fulfills a recommendation made by the NCI Clinical Trials Working Group (CTWG) to the National Cancer Advisory Board. CTRP is designed to be a comprehensive database of regularly updated information, including accrual, on all NCI supported clinical trials. This comprehensive database of the entire NCI portfolio will help identify gaps in clinical research and duplicative studies, as well as facilitate effective clinical trial prioritization, trial search and trial matching. The CTRP engineering works on system enhancements, software updates, and technical improvements to ensure that researchers and institutions can submit trial data efficiently, while ensuring compliance with regulatory standards. Additionally, the CTRP engineering team plays a key role in developing interfaces that allow for interoperability with other NCI systems and external databases, helping to streamline data sharing and foster collaboration.
- The CTRP PMO (Program Project Management Office) is responsible for overseeing and managing the CTRP at the NCI. Its primary function is to ensure that all NCI-

supported clinical trials are properly registered, updated, and tracked throughout their lifecycle. The PMO ensures the standardization and accuracy of clinical trial data, which is critical for research, regulatory compliance, and patient safety. It also provides support to researchers and institutions, helping them meet reporting requirements and ensuring the smooth submission of data. Additionally, the CTRP PMO manages the overall operations of the program, ensuring the system remains functional and up to date while enabling efficient data sharing and collaboration among the research community. Through these efforts, the CTRP PMO plays a key role in improving the transparency and efficiency of cancer clinical trials.

- The Clinical Trial Reporting Office (CTRO) provides task management, data abstraction, coding, characterization quality assurance, user support and reporting in support of NCI's Clinical Trials Reporting Program (CTRP). The CTRO manages CTRP content, including abstraction of submitted clinical trial protocols, updating and maintaining the content of the CTRP database, ensuring the accuracy of content through quality assurance procedures, managing the NCI's ClinicalTrials.gov account, and responding to queries from stakeholder in a timely and accurate manner.
- The semantic infrastructure (SI) of the NCI relies on well-defined vocabularies, terminologies, and data models to ensure that information from clinical trials, research studies, and healthcare systems can be easily shared and understood by different stakeholders. At the core of this infrastructure is the NCI Thesaurus, a large, curated vocabulary that provides consistent definitions for cancer-related terms, drugs, diseases, and treatments. Additionally, the NCI Metathesaurus integrates multiple biomedical terminologies to facilitate data interoperability across systems. This semantic foundation allows researchers, clinicians, and data scientists to collaborate more effectively by ensuring that data is represented consistently, enabling meaningful comparisons and analyses across studies. NCI's SI also supports key initiatives like the Cancer Data Standards Repository (caDSR), which provides common data elements (CDEs) and models to further standardize data collection and sharing, ultimately accelerating cancer research and improving patient outcomes.
- The Real-World Data (RWD) program at the NCI focuses on leveraging data generated from real-world settings, such as electronic health records (EHRs), cancer registries, claims data, and other healthcare sources, to enhance cancer research and improve patient outcomes. This program aims to complement traditional clinical trial data by providing insights from larger, more diverse patient populations and real-world clinical practices. By integrating and analyzing RWD, the NCI can better inform clinical decision-making, accelerate the development of new therapies, and improve the understanding of cancer progression in various healthcare settings. The program also supports data-sharing efforts and fosters collaboration between researchers, clinicians, and healthcare institutions to maximize the impact of real-world evidence in advancing cancer research and care.

The National Cancer Institute's Office of the Chief Information Officer (OCIO) serves as the central authority for enterprise-wide IT governance, strategic planning, and service delivery

across all NCI operations. Operating under federal IT management requirements including Federal Information Technology Acquisition Reform Act (FITARA), Federal Information Security Modernization Act (FISMA), and National Institute of Standards and Technology (NIST) guidelines, the OCIO oversees comprehensive IT infrastructure management, cybersecurity operations, technology standardization initiatives, and project management following established federal frameworks including Project Management Institute (PMI) standards and Office of Management and Budget (OMB) guidelines for IT capital planning.

The OCIO's core responsibilities include enterprise architecture planning aligned with the Federal Enterprise Architecture Framework; establishment and management of IT engineering standards consistent with NIH and HHS policies; implementation of IT security controls including continuous monitoring, risk assessment, and incident response; and management of IT support services encompassing help desk operations, asset management, and user provisioning across NCI's research and administrative environments.

OCIO delivers enterprise IT services that enable seamless user experiences across geographically distributed locations, maintain secure and high-performance computing environments supporting both administrative functions and computationally intensive research workloads, optimize service delivery through cost-effective solutions, and promote enterprise standardization while supporting specialized research requirements. This approach ensures IT services support NCI's cancer research mission while maintaining federal compliance with security, privacy, and operational requirements.

All contract awardees are required to utilize NIH-approved IT services and designated providers and collaborate effectively with the OCIO and its authorized representatives to execute technology-related efforts in full alignment with established enterprise requirements, standards, and operational expectations.

2.3. OBJECTIVE

NIH's mission is to seek fundamental knowledge about the nature and behavior of living systems and the application of that knowledge to enhance health, lengthen life, and reduce illness and disability.

To support, enable, and advance this mission, the goals of this IDIQ are the following:

1. Provide the NIH with access to high-quality contractor services that deliver results-oriented solutions to enhance biomedical informatics programs across the enterprise, including support to the NCI.
2. Strengthen the OCIO's ability to effectively manage, analyze, process, and share biomedical data in accordance with NIH/HHS policies, applicable federal regulations and scientific best practices.

To achieve these goals, the IDIQ will enable the provision of highly qualified contractor support with subject matter expertise in biomedical information technology systems, application and database design, development, implementation, and maintenance; data science; biomedical informatics; clinical and health data sharing policy implementation; and specialized

biomedical IT and informatics.

2.4. SCOPE

The Contractor shall provide several general categories of consulting, planning, process improvement, management, systems and software development, information technology operations, subject matter expertise, and terminology analysis, development, maintenance, publishing, and support services to accomplish the NIH mission.

These services shall include but not be limited to overall project and technical management, consulting, and technical support to the NIH/NCI, including participating in efforts supporting other NIH/NCI Divisions, Offices, and Centers in biomedical informatics, biomedical information technology, and cancer research projects in all stages of their life cycle in accordance with the tasks specified below. These services require interaction with multiple NIH/NCI DOC's, Government Program Manager (s), senior business and IT managers, program stakeholders, and other NIH contractors.

The scope of services to be provided shall fall under one or a combination of the following task areas:

1. Contract Management
2. Program and Project Management Support
3. Enterprise Architecture Services
4. Definition and Design Services
5. Development Services
6. Validation, Testing, and Quality Assurance
7. Implementation Management and Support
8. Semantics Support Services
9. Clinical Trials Reporting Office (CTRO) Support Services
10. Technical and Policy Consulting
11. Website Development and Maintenance, Content Management and Update
12. Data Science Services
13. Artificial Intelligence (AI) Services
14. Other IT Services
15. Transition Out Planning and Support

The Contractor shall report directly to the Contracting Officer's Representative (COR) and shall not play any role in directing or managing work of any other project or support Contractor staff. Work products shall include data files, information models, software design artifacts, software source code and executables, IT capacity (provisioned information systems, storage arrays, etc.) reports, recommendations, system and program requirements, analyses, management reports, and briefings.

2.5. TASK AREAS

Independently and not as an agent of the Government, the Contractor shall furnish all necessary

services, qualified personnel, material, equipment, and facilities, not otherwise provided by the Government, as needed to perform task assignments at the direction of the COR in the Task Areas specified below.

2.5.1. TASK AREA 1 - CONTRACT MANAGEMENT

The Contractor shall provide the technical and functional activities necessary for the management of Task Orders under this SOW and all work performed to meet SOW requirements. The Contractor shall ensure that personnel are sufficiently qualified to execute the tasks listed in this SOW. The Contractor shall assume responsibility to meet the cost, performance, and schedule requirements throughout the Contract execution.

Additionally, the Contractor shall participate in an IDIQ contract kickoff and management meetings, and provide monthly status reports monitoring the performance and progress toward successful completion of the Task Orders as applicable.

The Contractor shall ensure that all personnel complete appropriate training. This includes, but is not limited to, the use of training necessary for this contract and all applicable confidentiality, privacy, information security, and conflict of interest regulations and corresponding NIH, HHS and Federal policies and requirements. Contractor/subcontractor staff shall sign Non-Disclosure Agreements, if they are required to have access to data that contains personally identifiable or business sensitive and/or proprietary information.

2.5.1.1. Task Order Management Plan

The Contractor shall provide a Task Order Management Plan as specified in each Task Order outlining the tasks, sub tasks, and approaches to be used to provide the required services including the specific deliverable documents, analyses, and reports to be developed. When approved by the COR, this plan shall be used to monitor, control, and report progress, issues, and resource expenditure.

As part of each assigned Task Order, the Contractor shall provide the COR with a *Work Plan* for organizing, executing and monitoring the resource utilization, schedule, services and deliverables. This work plan when approved shall be incorporated into the *Task Order Management Plan*.

The Contractor shall support the development, maintenance, and document control/tracking, of project plans, roadmaps, protocol documents, spreadsheets, financial statements, and contract deliverable documents. The Contractor shall keep all documents secured in a system that allows folder/document/workspace management allowing for access to be granted to the Contractor, NCI, NIH, HHS, and any designated 3rd party.

Additionally, the Contractor shall collect and provide Government requested metrics and data that provide insight into the efficiency of processes needed to support this SOW; and provide recommendations for changes that improve work, management effectiveness, and end-user satisfaction and reduce costs.

2.5.1.2. Contractor Coordination and Interfaces

The Contractor shall provide mission critical services in a fast moving and highly integrated environment that requires effective communication and close coordination of plans, schedules, and activities. The Contractor shall interact with the NIH appointed COR; attend scheduled and ad hoc meetings; and interact with federal staff, stakeholders, other contractors and executive leadership. The Contractor shall establish and maintain effective and ethical customer relationships as befits a customer-focused service enterprise.

The Contractor and/or its subcontractors shall be required, as part of the performance of this effort, to work with other contractors working for the Government. Such other contractors shall not direct this Contractor and/or its subcontractors in any manner. Also, this Contractor and/or its subcontractors shall not direct the work of other contractors in any manner.

The Contractor shall work closely with other NIH/NCI contractors to implement procedures to ensure open, timely, and effective communication, information exchange, and reporting among the contractors and the Government. When necessary to ensure clarity and acceptance, these procedures shall be submitted to NIH/NCI management for review, comment, and direction and to ensure the final procedures are followed by all parties.

2.5.1.3. Status Meetings and Reports

As specified in the Task Order, the Contractor shall participate in recurring management meetings, provide scheduled status updates and monthly status and financial reports monitoring the performance, progress, and resource expenditures toward successful completion of the Task Order SOW. The Contractor shall identify and discuss known and anticipated changes to work schedules and costs in these meetings and seek approval from the Government prior to implementing those changes.

2.5.1.4. Contractor Quality Assurance/Quality Control Plan

Additionally, the Contractor shall provide and implement an approved *Quality Assurance Plan* as specified in the Task Order outlining how assigned task work efforts will be controlled to ensure development and delivery of high quality services and products; provide insights into work processes; and provide positive guidance to improve work efficiency and acceptance of results.

2.5.1.5. Contractor/Subcontractor Personnel

The Contractor shall be responsible for managing and overseeing the activities of all Contractor personnel, as well as subcontractor efforts used in performance of this effort. The Contractor shall ensure that personnel are sufficiently qualified to execute the tasks listed in this SOW. The Contractor's management responsibilities shall include all activities necessary

to ensure the accomplishment of required training and timely and effective support, performed in accordance with the requirements contained in the SOW. Resumes submitted for employees assigned to perform under the Task Order SOWs shall contain documented experience directly applicable to the functions to be performed. Further, these prior work experiences shall be specific and of sufficient variety and duration so that the employee is able to effectively and efficiently perform the functions assigned.

2.5.1.6. Transition-In Planning and Support

The Contractor shall provide a *Transition-In Plan* as specified in each Task Order outlining how it proposes to quickly and successfully provide qualified staff to implement the proposed technical approach, transition current and ongoing tasks, and activities, and ensure effective knowledge transfer from any outgoing contractor(s). After Contract award and receipt of government comments and suggested changes, this *Transition-In Plan* shall be finalized, delivered to the government, and implemented to ensure a timely and successful transition.

The *Transition-In Plan* shall provide a logical strategy and process to mitigate risk to the Government. The plan shall include the following as a minimum:

- Overview of how transition will be implemented and managed,
- Staffing plan and approach,
- Staffing plan implementation and management,
- Transition of key personnel and non-key personnel,
- Transition of existing efforts,
- Resources required for the transition,
- Roles and responsibilities,
- Timeline,
- Risks, issues, and associated implications, and
- Risk and issue mitigation processes and procedures.

Transition-in shall begin at Task Order award. The Contractor shall complete all transition-in efforts in accordance with the approved transition plan (to include the Government's validation of the Transition Plan, containing critical events, schedules, and approach) and be prepared to begin performance immediately.

2.5.2. TASK AREA 2 - PROGRAM and PROJECT MANAGEMENT SUPPORT

The Contractor shall provide a Program Manager to facilitate Government-Contractor communications. The Program Manager shall be fully assigned to this contract and shall be the primary technical and managerial interface between the Contractor, the Contracting Officer

(CO) and the Contracting Officer's Representative (COR). The name of the Program Manager shall be designated in writing to the CO and shall have full authority to act for the Contractor on all contract matters relating to daily operations. The Program Manager must have fluent ability to read, write, speak, and understand English.

The Contractor's Program Manager must be available during normal duty hours, as specified herein, and available to meet with Government personnel within 24 hours to discuss problems.

The Contractor's Program Manager shall meet with the CO/COR as necessary to maintain satisfactory performance and to resolve other issues pertaining to Government/Contractor procedures. At these meetings, a mutual effort will be made to resolve any and all problems identified. Action items stemming from these meetings shall be prepared by the Contractor, ownership assigned, and timeframe proposed, and furnished to the Government within two (2) workdays of the subject meeting.

The Contractor shall apply expert level knowledge and best program and project management practices to ensure effective management of activities and achievement of project cost, schedule, and quality goals. The Contractor shall follow and support NIH/NCI management policies and procedures and provide management documentation and products following NIH/NCI provided examples and templates. The contractor shall provide consulting and technical services supporting program and project management of the planning, acquisition, definition, design, development, implementation, operation, and maintenance of services, applications, and software systems and hardware and communications infrastructures to support biomedical research, informatics, and care delivery. The contractor shall apply expert level knowledge and best project management practices to ensure effective management of activities and achievement of project cost, schedule, and quality goals.

Services provided shall include, but not be limited to:

- Project planning, monitoring, and progress and resource utilization reporting,
- Developing project and program business cases, work breakdown structures management plans, effort and cost projections, and other required documentation,
- Examining alternatives for the potential projects and systems, non-compliant areas, risk areas related to project requirements and activities,
- Supporting project plan and business case updates, cost analyses updates, and cost tracking and reporting,
- Defining, documenting and aiding with milestone reviews, project schedules, tasks and assumptions,
- Supporting project governance processes and reviews,
- Performing technical and business requirements analysis and tracking artifacts and process for mapping to functionality and prioritization,
- Identifying and implementing project management tools to support project needs

- Defining, distributing, and maintaining project schedules and project team task lists,
- Maintaining project issues list and status, quality management, action item tracking, and monitoring risk management, and
- Providing consulting and training support to individual government project and task managers to help ensure successful, on-time, within budget project completion.
- Performing program and project assessment evaluations such as developing strategies and processes for evaluating progress and performance, developing metrics for success and processes for determining go-no-go decisions,
- Coordinating among teams and different NIH/NCI programs, and
- Developing and implementing communication strategies and mechanisms.
- Supporting data policy and process implementation for cancer research and clinical care initiatives
- Training and Knowledge Transfer Services including technical training development and delivery, digital literacy and skills development programs, knowledge management and documentation, and train-the-trainer programs

2.5.3. TASK AREA 3 – ENTERPRISE ARCHITECTURE SERVICES

The Contractor shall provide consulting and technical services to support the definition, design, development, implementation, operations, and management of an NIH-wide Enterprise and Information Architecture. This architecture shall include NIH/NCI operations as outlined in this SOW. The Contractor shall apply expert-level knowledge of Enterprise and Information Architecture, and business intelligence toolsets to support these efforts.

These services shall include, but are not limited to:

- Consulting services for NIH and NCI data repository, data architecture, and data warehousing initiatives,
- Support for the development of project and enterprise use cases,
- Enterprise Architecture Modeling support,
- Business process modeling services for current and future processes that drive NIH's investments in biomedical informatics and information technology,
- Program/project governance support, and
- Expertise in the development and maintenance of the NCI's Enterprise Conformance and Compliance Framework (ECCF) and its support artifacts.

2.5.4. TASK AREA 4 – DEFINITION AND DESIGN SERVICES

The Contractor shall provide consulting and technical services to support the definition and design of services, applications, and software systems used to support biomedical research,

informatics, and care delivery. The Contractor shall apply expert-level knowledge in information system design, Service-oriented Architecture, Cloud-based Architecture, system modeling and design, and other best practices.

The services shall include, but are not limited to:

- Business process modeling,
- Requirements gathering,
- Systems analysis,
- Level of effort, cost, and schedule analysis and validation,
- Data collection, modeling and analysis,
- Information modeling,
- System design, and
- User interface/User Experience (UI/UX) design.

This task shall involve assisting NIH and NCI in creating designs that are compatible with NCI system design criteria and compatibility guidelines. The Enterprise Conformance and Compliance Framework (ECCF) is the current conformance and compliance model used at CBIIT. The Contractor should be capable of adjusting the model based on other specified standards or frameworks in use for a specific effort.

2.5.5. TASK AREA 5 – DEVELOPMENT SERVICES

The Contractor shall provide biomedical information technology services to support state of the art software engineering and development of software and promote an understanding of biomedical research and care delivery policy requirements.

The Contractor shall provide staff with experience in the following technologies and processes: Agile (e.g., Kanban and Scrum), Cloud Infrastructure and Security, Development and Operations (DevOps), Python, serverless computing including but not limited to Amazon Web Services (AWS) Lambda, Cloud Managed Services (e.g., Quantum Ledger Database (QLDB), DynamoDB, Neptune, Aurora, RedShift), NoSQL, MongoDB, Security Assertion Markup Language (SAML), Java, Ruby, Oracle, Enterprise Architecture Modeling (e.g., Business Process Model and Notation (BPMN), ArchiMate), FISMA Compliance, AWS S3, AWS CloudFront, AWS Shield, AWS API Gateway, and other AWS services.

The Contractor shall support NIH activities that align with the NCI Cloud Computing Platform Management and Support (Attachment X), the NCI CBIIT Cloud Computing Platform Management Services Overview (Attachment X), and the NIH FISMA and FEDRAMP Information Security (Attachment X).

The Contractor shall design and implement iterative and/or agile software development lifecycle methodologies to achieve efficient delivery of small and high-value capabilities, rapid responses to changes in business priorities and requirements, and increase customer satisfaction.

These services shall include, but are not limited to:

- Software development,
- Database design, implementation, and administration,
- Software and data quality assurance and Independent Verification and Validation (IV&V) services,
- DevOps,
- Technical writing, and
- Development of training.
- In providing services supporting software engineering and development the Contractor shall:
 - Establish, document, and adhere to NIH/NCI software engineering best practices, software development, configuration management and monitoring standards per application, portfolio priorities and procedures,
 - Continually improve the NIH/NCI software solutions responding to evolving customer requirements and HHS / NIH / NCI policies, and the need to reduce the administrative burden on the user community,
 - Maintain product backlogs and incorporate desirable functionalities not included in early iterations into future releases,
 - Provide full life cycle software development support,
 - Manage and prioritize resources to meet the competing demands of software engineering service requests, and
 - Support containerization and cloud adoption with full-stack automation capabilities and migration of existing applications to NIH/NCI IT-managed cloud environments.

2.5.5.1. Scientific Application Development and Support

The Contractor shall provide full-life cycle software engineering activities for custom-built software application development to support automation of NIH/NCI Division/Office/Centers business and scientific workflows and processes. In addition, the Contractor shall provide the following custom-built application development and support services:

- Collaborate with NIH/NCI scientific system owners and business stakeholders to prepare and conduct application/solution demos, training sessions and participate in workgroup and committee meetings,
- Provide technical consultation to other NIH/NCI users whose systems leverage or depend on NIH/NCI custom-built scientific applications and/or solutions,
- Collaborate with Business Analysis team to provide dependable and responsive

production user support,

- Collaborate with Business Analysis and Quality Assurance/Testing teams to provide reliable operation and maintenance support, and
- Maintain, manage, enhance, and provide the integrity and security of databases, as necessary.

2.5.5.2. Scientific Application Hosting Services

The Contractor shall maintain and support an application hosting environment and provide optimal hosting solutions that meet the NIH/NCI application hosting requirements using on-premises and cloud-based resources. Contractor shall:

- Have a broad knowledge and technical capability to provide application hosting solutions for custom-built applications and commercial-off-the-shelf (COTS) products,
- Conduct necessary activities to ensure that the hosting environment and the hosted application systems conform to Government information security requirements, regulatory requirements, and NIH/NCI and project-specific requirements, as well as use the technologies in the Vendor Reading Library, and
- Proactively maintain hosting systems to comply with Government information security requirements, regulatory requirements, and NIH/NCI and project-specific requirements, and remediate any violations identified by the NIH/NCI Security Team.

2.5.5.3. C3D Application and Database Hosting Support

C3D (Cancer Central Clinical Database) is one of the NCI's clinical trials data management systems. It collects clinical trial data using standard case report forms (CRFs) based on common data elements (CDEs). C3D currently supports electronic submission of clinical trials data to the NCI's Clinical Data System (CDS) and the Clinical Trials Monitoring Service CTMS/Theradex). This system has been designed to integrate with NIH-wide and NCI information systems providing an array of informatics tools to support data standardization, validation, and quality assurance, as well as establishing security procedures to protect patient confidentiality and maintaining an audit trail to satisfy regulatory requirements.

The main technology of C3D is Oracle Clinical (OC), and the other components include: Jreview and NCI custom-built tools such as Data Extract, Data Loader, and Metadata Loader. C3D is one of the applications that is hosted in the CBIIT-managed hosting environment.

To be highly successful, the Contractor will be proficient in the main technology of C3D, Oracle Clinical (OC), including OC backend database, Oracle Weblogic Suite and web server, and OC front-end.

These required services include all tasks necessary to support C3D database and application hosting needs based on customer requirements in production tier. These services include, but are not limited to:

- Performing software installation and upgrades/patches including OC backend database, middle tier (Oracle Weblogic Suite), and OC front-end application,
- Providing operation and maintenance support for C3D. The major components include C3D database, middle-tier, OC front-end application, and C3D tools, and
- Performing tasks as specified in all other sections of Scientific Database Hosting Services in Section 5.5.2 above.

2.5.6. TASK AREA 6 – VALIDATION, TESTING, AND QUALITY ASSURANCE

The Contractor shall provide technical and consulting services to provide validation, testing, and quality assurance of software, applications, databases, web sites, and systems to meet NIH/NCI and other federal, state and local authority requirements. These services shall include but are not limited to:

- Development and implementation of formal software validation, testing, and quality assurance processes and the underlying Software Development Lifecycle (SDLC) processes required to implement validation processes,
- Development, implementation, and operation of validation processes and procedures to be used to assess compliance with compatibility guidelines, testing, and quality assurance requirements,
- Validation against the requirements of federal laws governing electronic information systems and protection of human subjects, including Health Insurance Portability and Accountability Act (HIPAA), 21 C.F.R. Part 11, and 45 C.F.R. 46 (usually referred to as “the Common Rule”),
- Testing and quality assurance of products including mapping functions to requirements, conducting testing activities, reporting test results, issues, and risks, and recommending mediation and/or resolution of problems identified,
- Support for submissions to external validation and certification bodies, and
- Development of supporting validation, testing, and quality assurance documentation required by developers, stakeholders, and user communities.

2.5.7. TASK AREA 7 – IMPLEMENTATION MANAGEMENT AND SUPPORT

The Contractor shall provide services to support the implementation management of NIH/NCI biomedical research, and NIH/NCI business systems. These services include, but are not limited to:

- Support for the planning, management, tracking, and reporting on deployment of information systems at NIH, NCI, CBIIT leased, and NIH/NCI supported sites, as well as in the cloud,
- Database management services including planning, evaluation, risk and issue analysis

and management support,

- Cloud Operations, Cloud General Security System, and Cybersecurity services, and
- Data analysis and reporting services include utilization, effectiveness, acceptance, performance, risks, issues, etc.
- IT Operations and Infrastructure Support including system monitoring, performance optimization, capacity planning, incident and problem management, and enterprise platform administration for Software as a Service (SaaS) solutions (ServiceNow, M365, enterprise applications)
- Information Technology Service Management (ITSM) including Information Technology Infrastructure Library (ITIL) process implementation and improvement, service catalog management, change and configuration management, and service level management.

In providing services supporting software engineering and development the Contractor shall:

- Provide operational and maintenance support for software solutions deployed to production such as:
 - User production support,
 - Incident triage and resolution,
 - Feature request routing and tracking,
 - Application performance monitoring, and
 - System reliability.
- Develop and maintain technical documentation including system, training, and user guides for software solutions deployed to production,
- Assist NIH/NCI to attain and maintain the Authorization to Operate (ATO) by:
 - Developing the initial security startup documentation early in the development lifecycle including the Federal Information Processing Standards FIPS-199, the Privacy Impact Assessment (PIA), Business Impact Assessment (BIA) and the Electronic Authorization Forms,
 - Working with the infrastructure team to develop and maintain the System Security Plan (SSP) and all the other required security documentation as defined in the NIH Security Handbook,
 - Updating the SSP and related documentation when significant changes are made to the architecture or security posture of any application. This includes providing a Security Impact Assessment that outlines all the changes, and
 - Supporting the Information Systems Assessment and Authorization (SA&A) Team and performing software engineering security remediation when necessary.
 - Develop and implement data analytics and reporting solutions using NIH/NCI-approved business intelligence tools and processes.

2.5.8. TASK AREA 8 – SEMANTICS SUPPORT SERVICES

The Contractor shall provide technical and consulting support for the management of semantic operations that are a major component of NCI software development processes. These services shall include, but are not limited to, assistance in the management and implementation of the following:

- Creation, deployment, and maintenance of Enterprise Vocabulary Services, including controlled terminologies, metadata, vocabularies, and ontologies for the support of NCI operations and software development activities,
- Creation, deployment, and maintenance of Common Data Elements for the support of NCI operations and software development activities in the NCI's cancer Data Standards Repository (caDSR) and any successor systems,

Creation, deployment, and maintenance of additional semantic services as required by NCI.

2.5.9. TASK AREA 9 – CLINICAL TRIALS REPORTING OFFICE (CTRO) SUPPORT SERVICES

- NCI, including APIs to semantic services.

The Contractor shall provide project management, data management, and reporting services, e.g., data abstraction, quality assurance, user support and reporting, in support of the NCI Clinical Trials Reporting Program (CTRP.)

The Contractor shall provide services that include, but are not limited to, the following:

- Provide accurate and consistent abstraction, curation and coding of clinical trial administrative and scientific data,
- Abstract new trial and amendment submissions, implement changes in trial status and participating sites,
- Manage updates to submissions; ensure automated solicitations are scheduled and sent to receive trial updates twice a year,
- Manage the reporting of clinical trial accrual into CTRP, ensuring the timeliness of reporting and compliance with accrual data reporting guidelines,
- Develop and implement a continuous quality assurance plan to assure accuracy and consistency of data entered and maintained in CTRP,
- Implement, maintain, and update Standing Operating Procedures (SOPs); identify departures from SOPs; recommend and implement appropriate changes in a timely manner,
- Maintain a list of operational issues; seek guidance from the Contracting Officer's Representative (COR), Government lead, and/or designee on how to prioritize the resolution of operational issues; work with the COR, Government lead, and/or designee to develop a timeline to resolve operational issues,

- Meet with staff in NCI's Coordinating Center for Clinical Trials (CCCT), other Contractor staff, and internal and/or external NCI stakeholders upon request, to discuss topics, such as operational issues, technical issues and/or other related concerns,
- Manage the NCI's ClinicalTrials.gov Protocol Registration and Results System (PRS) account for NCI Sponsored trials registered in CTRP,
- Manage compliance with results reporting for NCI sponsored trials and other trials directly funded by NCI, e.g., trials managed by NCI's Cancer Therapy Evaluation Program (CTEP),
- Provide user support, e.g., responding to incoming queries, sending messages regarding CTRP as directed to listservs, developing training materials regarding CTRP and providing information webinars and/or presentations upon request,
- Manage the accuracy and curation of personnel and organizations associated with clinical trials,
- Provide workload, status, performance, data quality, and other reports upon request from the COR, Government lead, and/or designee, and
- Develop training and documentation materials, e.g., PowerPoint presentations, website content, and updates for CTRO related activities.

2.5.10. TASK AREA 10 – TECHNICAL AND POLICY CONSULTING

The Contractor shall provide subject matter expertise and support services in the task areas delineated in the scope section of this SOW, as requested by NIH/NCI to support required activities, tasks, projects, and initiatives. The expertise and support provided shall assist Federal staff to plan, implement, control, and manage on-going and new work to achieve approved NIH/NCI plans, goals, and objectives.

This includes executive and strategic support services such as CIO office strategic planning support, IT governance and oversight assistance, executive briefings and reporting, strategic initiative planning and coordination, and training and knowledge transfer services including technical training development and delivery, digital literacy and skills development programs, knowledge management and documentation, and train-the-trainer programs, all as directed by the OCIO and in coordination with established enterprise requirements.

2.5.11. TASK AREA 11 – WEBSITE DEVELOPMENT AND MAINTENANCE, CONTENT MANAGEMENT, AND UPDATE

The Contractor shall provide high-quality integrated digital services and product development expertise to ensure compliance with NIH communications guidelines, policies, and direction. The Contractor shall demonstrate a clear understanding of integrated digital strategy development, implementation, and management to communicate cancer data science and informatics information to multiple audiences and will effectively interact with NIH/NCI

Communications Team so that new web properties properly interact and reflect CBIIT and NCI's web presence standards.

The Contractor shall provide staff with technical skills and expertise to assist NIH/NCI to maintain and update NIH/NCI and project web sites. This support shall include, but are not limited to the following:

- Developing creative briefs/reports that outline the business goals, audience needs, problems that need to be solved, and a general approach for each major web project,
- Providing support and subject matter expertise to leverage existing NIH channels to ensure websites have documented user needs that are quantitative or qualitative in order to determine effective and efficient use of resources and user benefit. Support shall include discovery through user interviews, voice of the customer surveys, A/B testing, usability testing, and other relevant approaches,
- Submitting requests for project approvals through the NIH New Website/Web Application Process,
- Ensuring that websites/web applications incorporate appropriate analytics codes to support web analytics Federal reporting requirements,
- Developing measurement plans and key performance indicators for products and projects, creating campaigns and monitoring and reporting progress toward accomplishing goals of the campaigns as specified in the measurement plans,
- Maintaining sites including updating technologies and links as required,
- Keeping site content current through loading updated content provided,
- Ensuring all digital assets align with NIH branding and identity and guidelines,
- Conducting and supporting audience research and the use of analytics to enhance the user experience with NIH/NCI websites, web applications, and social media posts,
- Providing appropriate support to NIH/NCI to obtain OMB clearance where required, and
- Providing development and programming support as needed to ensure all products are:
 - Audience-focused,
 - Data-informed,
 - Collaborative, and
 - Agile processes.

The Contractor shall support NIH/NCI activities in a manner that meets the requirements of the 21st Century Integrated Digital Experience Act (IDEA) and OMB Memo 23-22, OMB regulations such as the Paperwork Reduction Act, and HHS policies and guidelines related to digital communications, and other federal and NIH requirements which include Section 508 of the Rehabilitation Act, the Digital Analytics Program, the Plain Writing Act, as well as security mandates.

2.5.12. TASK AREA 12 – DATA SCIENCE SERVICES

The Contractor shall provide consulting and technical services to support data science initiatives that enhance NIH/NCI's ability to collect, process, integrate, and analyze biomedical data in support of cancer research and clinical care. The Contractor shall apply expert-level knowledge in data management, data engineering, cloud computing, and statistical analysis to ensure the effective use of data resources across NIH/NCI.

The services shall include, but are not limited to:

- Data Management and Standardization
 - Developing and implementing data management strategies to support the FAIR (Findable, Accessible, Interoperable, and Reusable) principles,
 - Supporting data governance, metadata management, and data harmonization across research programs,
 - Maintaining and enhancing Common Data Elements (CDEs) and ontologies for cancer research, and
 - Semantics Infrastructure operations support including terminology and CDE development, maintenance as well as providing interfaces to SI resources to be consumed.
- Biomedical Data Integration and Engineering
 - Designing and maintaining data pipelines, Extract, Transform, and Load (ETL) processes, and Application Programming Interfaces (APIs) to integrate structured and unstructured biomedical datasets,
 - Supporting the development of data repositories and databases, ensuring compliance with interoperability standards such as FHIR, HL7, and caDSR, and
 - Enhancing NCI's Cancer Research Data Commons (CRDC) infrastructure and supporting integration with other NIH data resources.
- Data Analytics and Visualization
 - Developing data dashboards, reports, and analytics platforms to enable data-driven decision-making,
 - Supporting exploratory data analysis and trend analysis for clinical trials, genomics, proteomics, clinical, wearable devices, and imaging data, and
 - Implementing advanced data visualization techniques to support research insights.
- High-Performance Computing & Cloud-Based Data Processing
 - Utilizing cloud computing (Amazon Web Services (AWS), Azure, Google Cloud) for large-scale biomedical data processing,
 - Implementing distributed computing and parallel processing techniques to accelerate data analysis, and
 - Supporting the integration of data lakes, federated data access, and cloud-based

analytical environments.

- Data Security, Privacy, and Compliance
 - Ensuring compliance with NIH, HHS, and federal data security regulations, including but not limited to FISMA, HIPAA, 21 Code of Federal Regulations CFR Part 11, and NIH-level Identity Proofing Requirements,
 - Implementing data anonymization, encryption, and secure access controls for sensitive research datasets, and
 - Supporting data audit, provenance tracking, and policy enforcement for data-sharing initiatives.
- Large Scale Data Harmonization and Phenotyping
 - Developing Cancer Phenotypes in National Clinical Cohort Collaborative (N3C), National Patient-Centered Clinical Research Network (PCORNet), Observational Health Data Sciences and Informatics (OHDSI) Observational Medical Outcomes Partnership (OMOP), and other large-scale network and initiatives,
 - Supporting data harmonization and quality checks to ensure data is represented properly,
 - In coordination with the subject matter experts, implementing predictive or other models to study data (linked or available in the network) for scientific development, and
 - Developing libraries and codes to analyze and clean to make read for scientific discoveries.

2.5.13. TASK AREA 13 – ARTIFICIAL INTELLIGENCE (AI) SERVICES

The Contractor shall provide consulting, research, and technical support to develop and integrate Artificial Intelligence (AI) and Machine Learning (ML) capabilities into NCI's biomedical informatics and research initiatives. The Contractor shall apply expert-level knowledge in AI algorithms, deep learning, Natural Language Processing (NLP), and AI model governance to support data-driven cancer research and clinical applications.

The services shall include, but are not limited to:

- AI-Driven Biomedical Imaging and Radiomics
 - Developing AI models for automated cancer detection, tumor classification, and radiomic feature extraction from medical imaging data,
 - Implementing deep learning techniques for histopathology, radiology, and microscopy image analysis, and
 - Supporting AI-based radiomics pipelines that integrate with existing imaging repositories and clinical decision-support systems.
- AI-Powered Genomic and Precision Medicine Applications
 - Applying AI models for genomic variant classification, biomarker discovery, and

- drug response prediction,
 - Supporting AI-driven integration of multi-omics data (genomics, proteomics, transcriptomics) for precision oncology, and
 - Developing AI-powered platforms for personalized treatment recommendations and risk stratification.
- Natural Language Processing (NLP) and Large Language Models (LLMs) for Clinical Data
 - Developing NLP and LLM pipelines to extract key insights from unstructured clinical and scientific text,
 - Supporting AI-based clinical trial matching to help automate patient recruitment and eligibility screening. Implementing data quality measures to improve availability, reusability, and quality of data for research purposes,
 - Utilizing AI models for safety events detection (e.g. adverse events, etc.) and capture of necessary and supporting data,
 - Implementing AI-powered knowledge graphs to enhance semantic search and biomedical literature analysis,
 - Developing and supporting pipelines that generate consistent and reliable plain language and patient-facing information (e.g., informed consent, research study abstract, recruitment documents, etc.),
 - Utilizing LLMs to create and update information repositories containing federal regulations and guidance associated with specific topics (e.g., responsible and ethical use of AI in research and healthcare),
 - Implementing multiagentic approaches to enable information management and natural language queries at multiple levels, across research programs and disciplines, and federated across Agencies, and
 - Developing methods for dynamic tracking and monitoring the availability of AI models (e.g., training information, tokenization, model performance, best uses, where to access, approved for federal use, etc.). Developing pipelines for streamlined and realistic generation of synthetic clinical data.
- AI Model Development, Training, and Deployment
 - Designing, training, and validating machine learning and deep learning models for biomedical research,
 - Developing AI predictive models for cancer progression, therapeutic response, and patient stratification.
 - Applying deep learning for imaging data, including spatial transcriptomics and histopathology in oncology.
 - Implementing AI model pipelines using trusted AI frameworks (e.g., TensorFlow, PyTorch, Scikit-learn, etc.),
 - Implementing AI-driven drug discovery and repurposing using integrative multi-

- omics cancer datasets, and
 - Supporting AI model monitoring, re-training, and drift detection for continuous improvement.
- AI Governance, Ethics, and Explainability
 - Ensuring transparency, fairness, and accountability in AI models, including bias detection and mitigation strategies,
 - Supporting compliance with federal regulations, Executive Orders, NIST AI risk management frameworks, and NIH AI policies,
 - Implementing Explainable AI (XAI) techniques to provide interpretability and trustworthiness in AI model development and research, and
 - Providing model cards or similar artifacts to represent AI model details and expectations.
- AI for Data Science and Decision Support
 - Applying predictive analytics and AI-driven decision support systems for clinical and research applications,
 - Supporting the use of AI for data harmonization, knowledge discovery, and automated metadata generation,
 - Developing AI-powered tools for automated data annotation, labeling, and quality assessment.
- AI Infrastructure and Cloud-Based AI Model Training
 - Leveraging high-performance computing (HPC) and cloud-based AI resources for scalable model training,
 - Implementing AI model deployment and inference optimization in cloud environments such as AWS, Azure, and Google Cloud, and
 - Supporting federated learning, edge AI or other frameworks for collaborative AI model (re-)training approaches.
- AI Privacy and Model Security
 - Utilizing zero-retention based models for sensitive, PII/PHI data, and
 - Ensuring access to data repositories containing PII/PHI through private links.

2.5.14. TASK AREA 14 – OTHER SERVICES REQUIRED TO SUPPORT SOW REQUIREMENTS

The Contractor shall provide other technical and systems services in support of NIH/NCI tasks and projects within the scope of this SOW as required and directed by the Government's assigned COR.

2.5.15. TASK AREA 15 – TRANSITION OUT PLANNING AND SUPPORT

In the final year of Task Order performance, or in the event the Government requirements necessitate a re-compete, the Contractor shall develop and deliver *Transition-Out Plans* prior to contract expiration, or termination. The purpose of the *Transition-Out Plan* is to ensure an orderly and successful transition of contract functions, should the incumbent Contractor not be selected during the re-competition of the award.

This *Transition-Out Plan* shall include processes, procedures, and activities the Contractor shall use, if not selected for a follow-on Task Order, to adequately educate, train and transfer knowledge to the successful contractor's employees and smoothly and effectively transition work activities to the new Contractor. When tasked, the Contractor shall implement the Transition Plan and assist the NIH/NCI to transition knowledge, processes, tasks, activities, etc. to the follow-on contractor.

The plan shall detail transition-out activities to assure continuity of operations and the execution of a smooth and timely transition. Transition-out activities shall be coordinated through the COR and/or other designated federal staff.